# COM 5506
# Social Network Analysis for Communication

*L4 Small World Networks*

SEM B 2025/2026

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# Idealistic models

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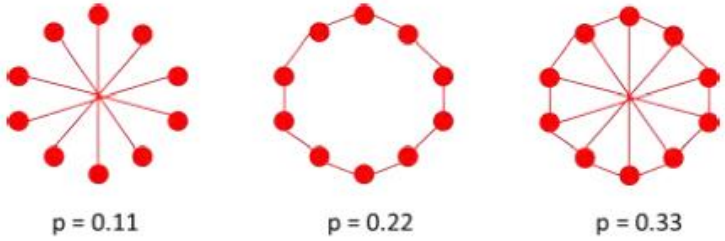
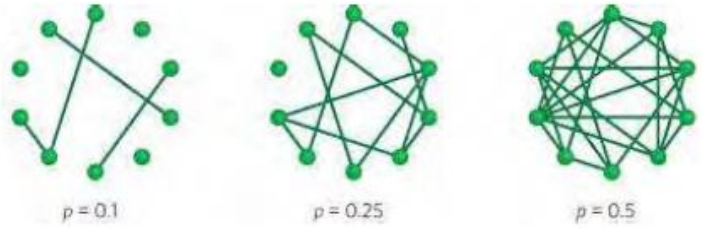
Euler (1735):

- Regular graphs
- The probability ($p$) for all edges in a graph is *uniform*




Erdos and Renyi (1959)

- Random graphs
- $p$ varies across edges based on a *Poisson* distribution




- Both types of network are a good starting point for understanding network structure, but both are too extreme to exist in human societies.

# The Milgram Experiment (1967)

- Several hundred people in Omaha, Nebraska, were asked to mail a letter to a friend who needed to mail the letter to another friend until it reaches a specific stockbroker in Boston.
- About one-third of the letters finally arrived at the destination with an average steps of 6.5.

# Six degrees of separation

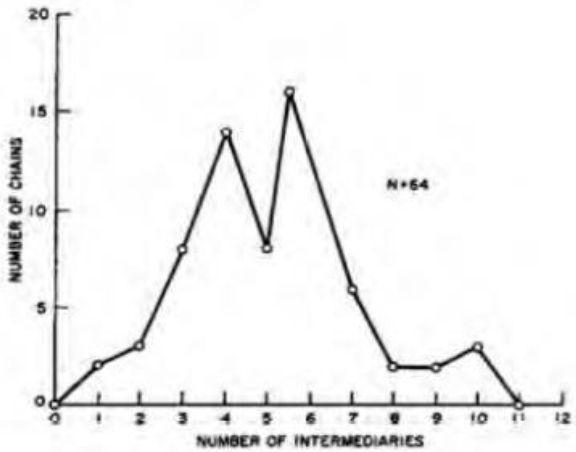



Figure 2.10: A histogram from Travers and Milgram's paper on their small-world experiment [391]. For each possible length (labeled "number of intermediaries" on the $x$-axis), the plot shows the number of successfully completed chains of that length. In total, 64 chains reached the target person, with a median length of six.

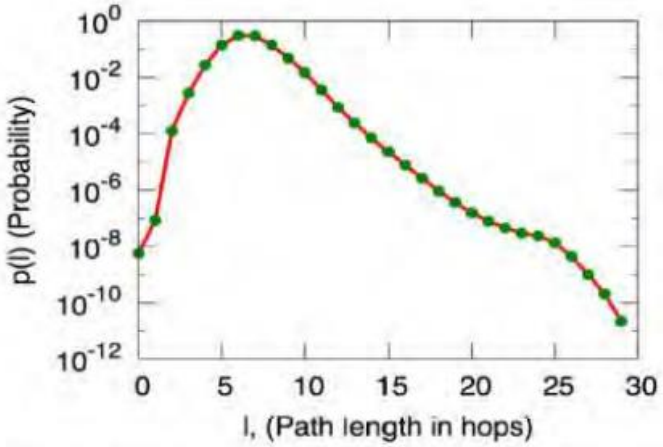



Figure 2.11: The distribution of distances in the graph of all active Microsoft Instant Messenger user accounts, with an edge joining two users if they communicated at least once during a month-long observation period [273].

- What is the metric of small world? The longest (i.e., diameter), the average, or the shortest distance?

# Email experiment

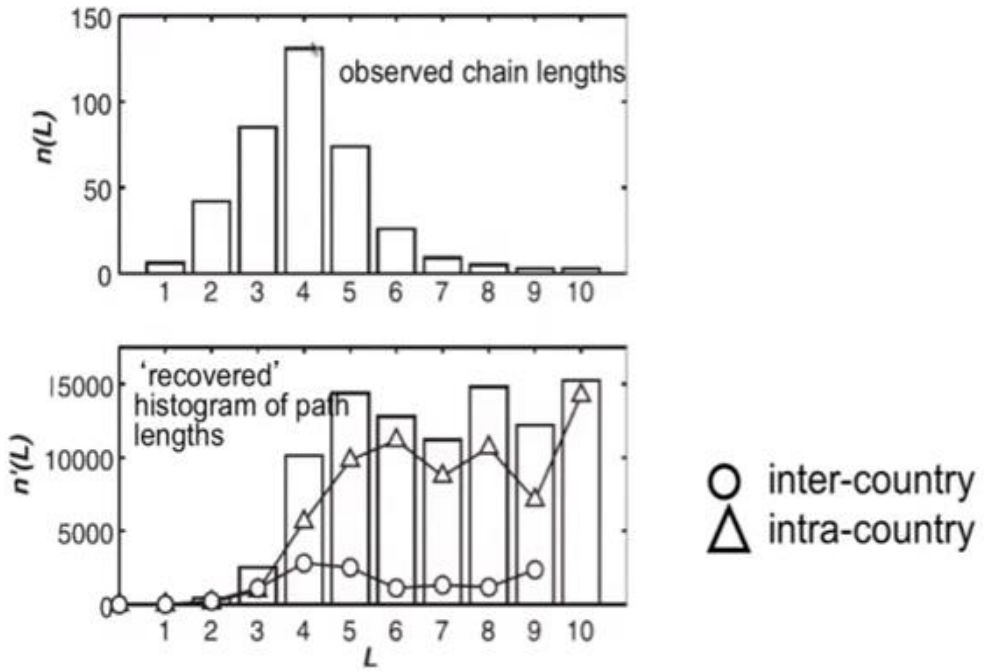
- Dodds, Muhamad, Watss, (2003) Science 301

- 18 targets
- 13 different countries

- 60000+ participants
- 384 reached their targets
- Average path length 4.0

# Attrition rate and accuracy

- Uncompleted chains and distance
- What bias is introduced by uncompleted chains?
- Are longer or shorter chains more likely to be completed?
- If each intermediate person in the chain has 0.5 probability of passing the letter on, what is the likelihood of a chain being completed of length 5?




Source: An Experimental Study of Search in Global Social Networks: Peter Sheridan Dodds, Roby Muhamad, and Duncan J. Watts (8 August 2003); Science 301 (5634), 827.

# Navigation and accuracy

- Individual do not have the knowledge about the global structure of a network.
- Do people find the shortest paths?
- Killworth, McCarty, Bernard & House (2005)
  - Find less than optimal choice for next link in chain is made ½ of the time

# Webpage links

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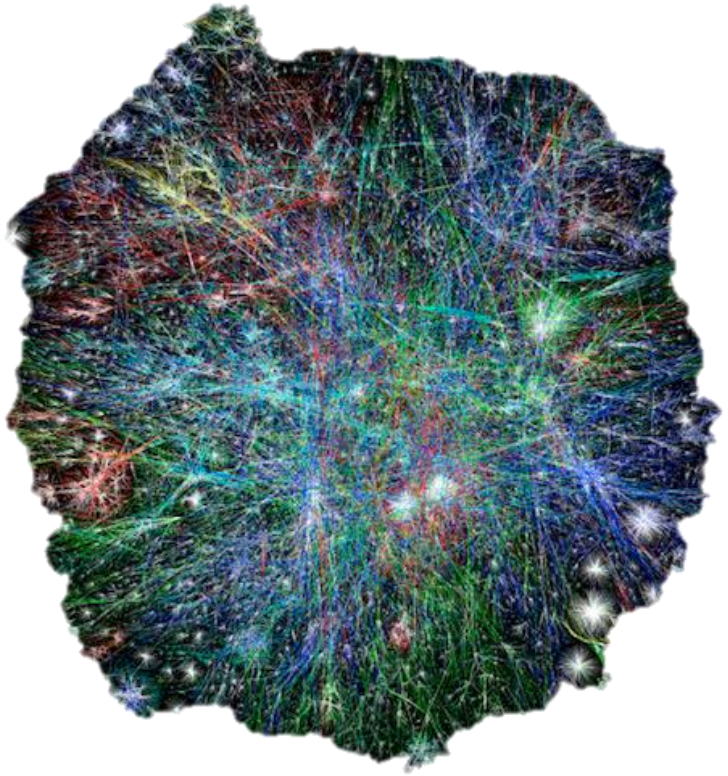
## Any Two Pages on the Web Are Connected By 19 Clicks or Less

There are more than 14 billion pages on the web, but they are linked by hyperconnected nodes, like Hollywood actors connected through Kevin Bacon

By Joseph Stromberg
SMITHSONIAN.COM
FEBRUARY 18, 2013



# Facebook makes the world even smaller

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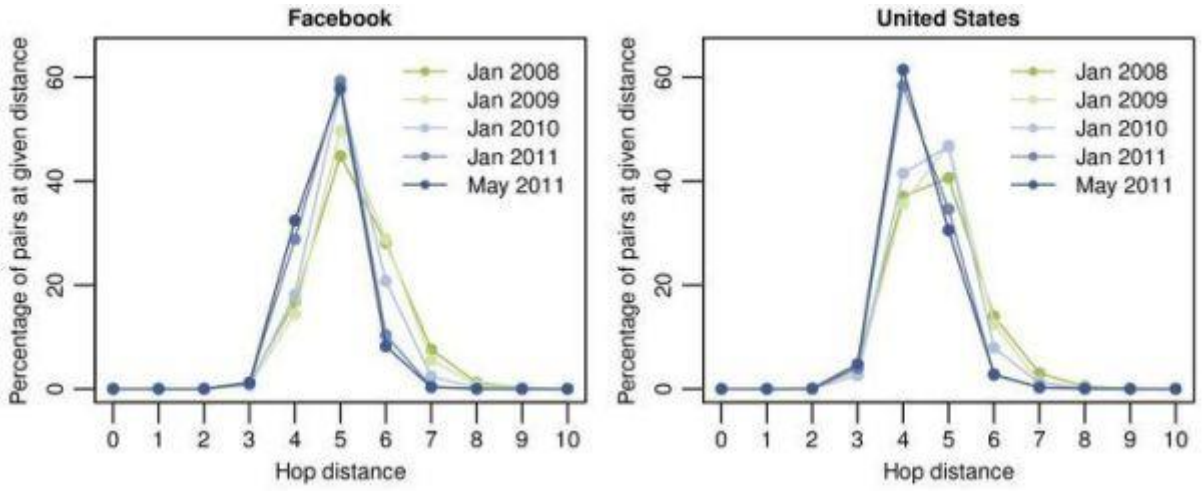
- An analysis of 700 million users on Facebook found that the average path length between all pairs of friend is 4.74, or a 5 degree of separation.



# Small-world networks are everywhere

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| | Network | Type | N of Nodes $n$ | N of Edges $m$ | Mean Degree $z$ | Mean Distance $\ell$ | $\alpha$ | $C^{(1)}$ | $C^{(2)}$ | $r$ | Ref(s). |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Social | film actors | undirected | 449 913 | 25 516 482 | 113.43 | 3.48 | 2.3 | 0.20 | 0.78 | 0.208 | [20, 415] |
| | company directors | undirected | 7 673 | 55 392 | 14.44 | 4.60 | – | 0.59 | 0.88 | 0.276 | [105, 322] |
| | math coauthorship | undirected | 253 339 | 496 489 | 3.92 | 7.57 | – | 0.15 | 0.34 | 0.120 | [107, 181] |
| | physics coauthorship | undirected | 52 909 | 245 300 | 9.27 | 6.19 | – | 0.45 | 0.56 | 0.363 | [310, 312] |
| | biology coauthorship | undirected | 1 520 251 | 11 803 064 | 15.53 | 4.92 | – | 0.088 | 0.60 | 0.127 | [310, 312] |
| | telephone call graph | undirected | 47 000 000 | 80 000 000 | 3.16 | | 2.1 | | | | [8, 9] |
| | email messages | directed | 59 912 | 86 300 | 1.44 | 4.95 | 1.5/2.0 | | 0.16 | | [136] |
| | email address books | directed | 16 881 | 57 029 | 3.38 | 5.22 | – | 0.17 | 0.13 | 0.092 | [320] |
| | student relationships | undirected | 573 | 477 | 1.66 | 16.01 | – | 0.005 | 0.001 | −0.029 | [45] |
| | sexual contacts | undirected | 2 810 | | | | 3.2 | | | | [264, 265] |
| Information | WWW nd.edu | directed | 269 504 | 1 497 135 | 5.55 | 11.27 | 2.1/2.4 | 0.11 | 0.29 | −0.067 | [14, 34] |
| | WWW Altavista | directed | 203 549 046 | 2 130 000 000 | 10.46 | 16.18 | 2.1/2.7 | | | | [74] |
| | citation network | directed | 783 339 | 6 716 198 | 8.57 | | 3.0/– | | | | [350] |
| | Roget's Thesaurus | directed | 1 022 | 5 103 | 4.99 | 4.87 | – | 0.13 | 0.15 | 0.157 | [243] |
| | word co-occurrence | undirected | 460 902 | 17 000 000 | 70.13 | | 2.7 | | 0.44 | | [119, 157] |
| Technological | Internet | undirected | 10 697 | 31 992 | 5.98 | 3.31 | 2.5 | 0.035 | 0.39 | −0.189 | [86, 148] |
| | power grid | undirected | 4 941 | 6 594 | 2.67 | 18.99 | – | 0.10 | 0.080 | −0.003 | [415] |
| | train routes | undirected | 587 | 19 603 | 66.79 | 2.16 | – | | 0.69 | −0.033 | [365] |
| | software packages | directed | 1 439 | 1 723 | 1.20 | 2.42 | 1.6/1.4 | 0.070 | 0.082 | −0.016 | [317] |
| | software classes | directed | 1 377 | 2 213 | 1.61 | 1.51 | – | 0.033 | 0.012 | −0.119 | [394] |
| | electronic circuits | undirected | 24 097 | 53 248 | 4.34 | 11.05 | 3.0 | 0.010 | 0.030 | −0.154 | [155] |
| | peer-to-peer network | undirected | 880 | 1 296 | 1.47 | 4.28 | 2.1 | 0.012 | 0.011 | −0.366 | [6, 353] |
| Biological | metabolic network | undirected | 765 | 3 686 | 9.64 | 2.56 | 2.2 | 0.090 | 0.67 | −0.240 | [213] |
| | protein interactions | undirected | 2 115 | 2 240 | 2.12 | 6.80 | 2.4 | 0.072 | 0.071 | −0.156 | [211] |
| | marine food web | directed | 135 | 598 | 4.43 | 2.05 | – | 0.16 | 0.23 | −0.263 | [203] |
| | freshwater food web | directed | 92 | 997 | 10.84 | 1.90 | – | 0.40 | 0.48 | −0.326 | [271] |
| | neural network | directed | 307 | 2 359 | 7.68 | 3.97 | | 0.18 | 0.28 | −0.226 | [415, 420] |

source:

M.E. Newman 2003

# Try the small world

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- Start from City U home page, click a link you think most likely to take you to final destination, and follow the path step by step.
- Do not use search engine during the whole process.
- Count the number of steps between.

- Final destination: Prof. Zhang Hongzhong. School of Journalism and Communication, Beijing Normal University.

# Why the world is so small?

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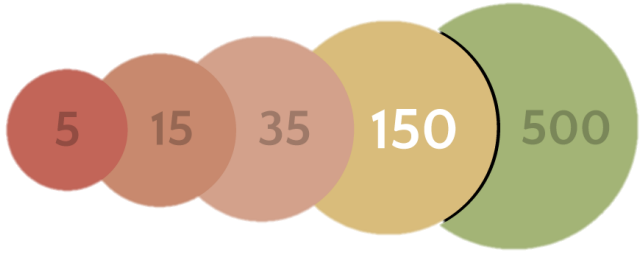
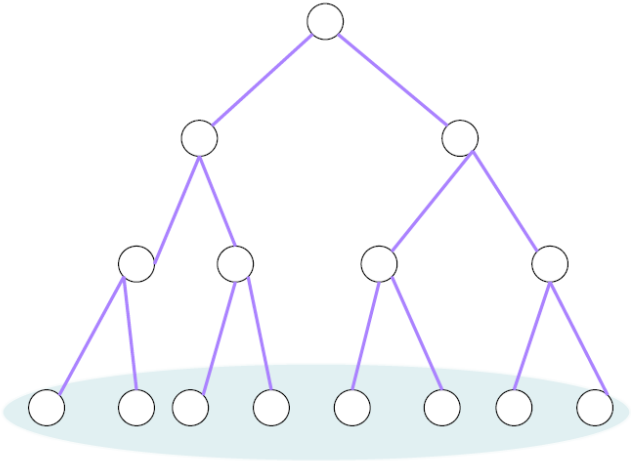
- Dunbar's number
  - Robin Dunbar (1992): humans can only comfortably maintain 150 stable relationships




- Primary groups
  - Family member 5
  - Relatives 10
  - Buddies 20
  - Other friends 115
- Secondary groups
  - Acquaintances 500-1000

# Question

- Ignore the fact that many of your friends' friends are your friends as well. If everyone has 500 friends, the average person would have how many friends of friends?



# If network is completely random

- You 1
- Your friend 500
- Your friend's friend 250000
- Your friend$^3$ 125000000
- Your friend$^4$ 62500000000
- (almost 10 times of current population on the planet)

7.4 billion

In demographics, the world population is the total number of humans currently living. As of March 2016 , it was estimated at **7.4 billion**, an all-time record high. The United Nations estimates it will further increase to 11.2 billion in the year 2100.

# If network is completely cliquish

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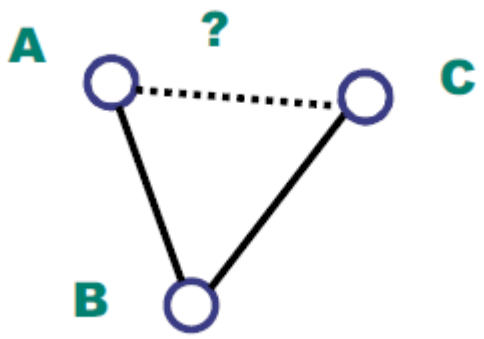
If the network is completely cliquish, that is all your friends of friends were also directly your friends.
Then you would be a part of an isolated clique.

# Transitivity

- My friends' friends are likely to be my friends




- How to quantify that? To count the number of closed triplets in a network!

# Clustering

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- Global clustering coefficient –

$$C = \frac{3 \times \text{number of triangles}}{\text{number of connected triplets of vertices}} = \frac{\text{number of closed triplets}}{\text{number of connected triplets of vertices}}.$$

A triplet consists of three connected nodes. A triangle therefore includes three closed triplets, one centered on each of the nodes.

# Clustering

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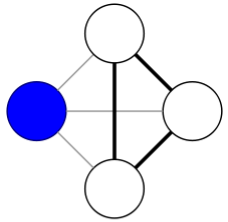
- Local clustering coeffcient (Watss & Strogatz, 1998)
- Duncan J. Watts and Steven Strogatz introduced the measure in 1998 to determine whether a graph is a small-world network.

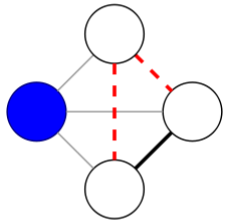
$$C_i = \frac{\text{\# of connections between i's neighbors}}{\text{max \# of possible connections between i's neighbors}}$$

$$C_{i\ \text{directed}} = \frac{\text{\# directed connections between i's neighbors}}{n_i * (n_i - 1)}$$

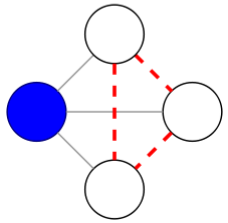
$$C_{i\ \text{undirected}} = \frac{\text{\# undirected connections between i's neighbors}}{n_i * (n_i - 1)/2}$$



c = 1



c = 1/3



c = 0

# Clustering

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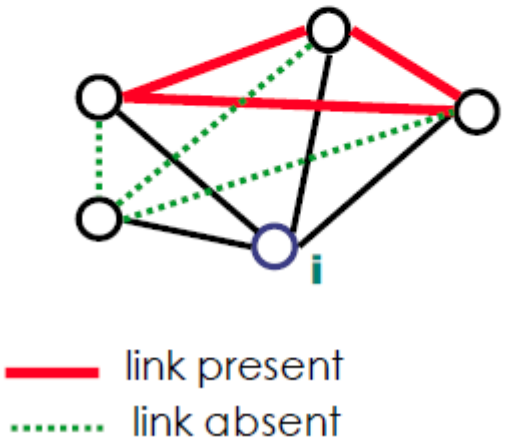
- Average over all n vertices

$$C = \frac{1}{n}\sum_i C_i$$




$n_i = 4$
max number of connections:
$4*3/2 = 6$
**3** connections present
$C_i = 3/6 = 0.5$

# Criteria of small world network

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- Two features
  - High clustering
  - Low average shortest path

$$C_{\text{network}} >> C_{\text{random graph}}$$

$$l_{\text{network}} \approx \ln(N)$$

# Watts-Strogatz model: Generating small world graphs

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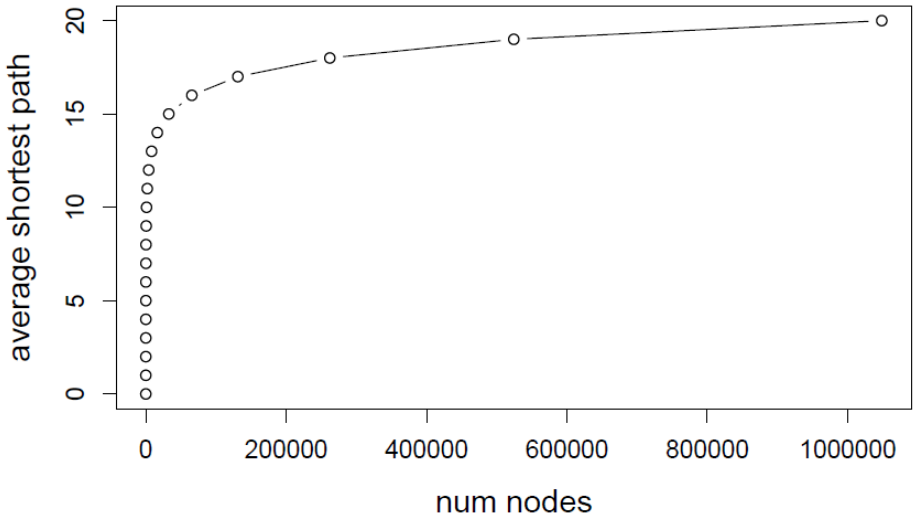
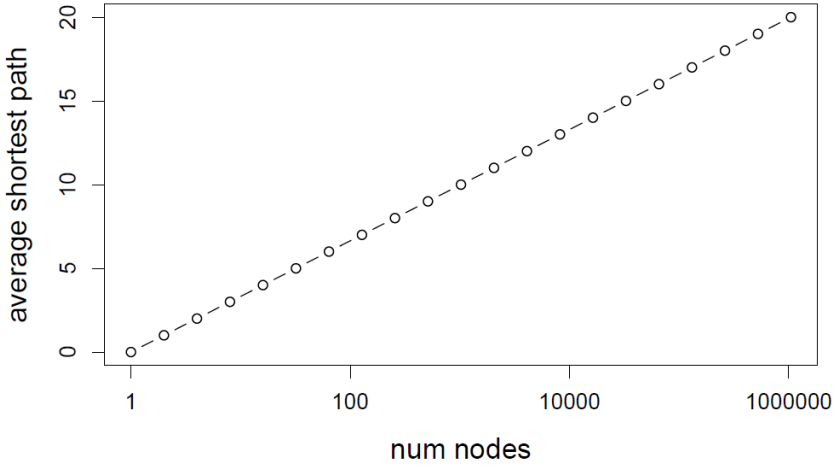
Select a fraction p of edges
Reposition on of their endpoints

Add a fraction p of additional edges leaving underlying lattice intact

# Erdos-Renyi model

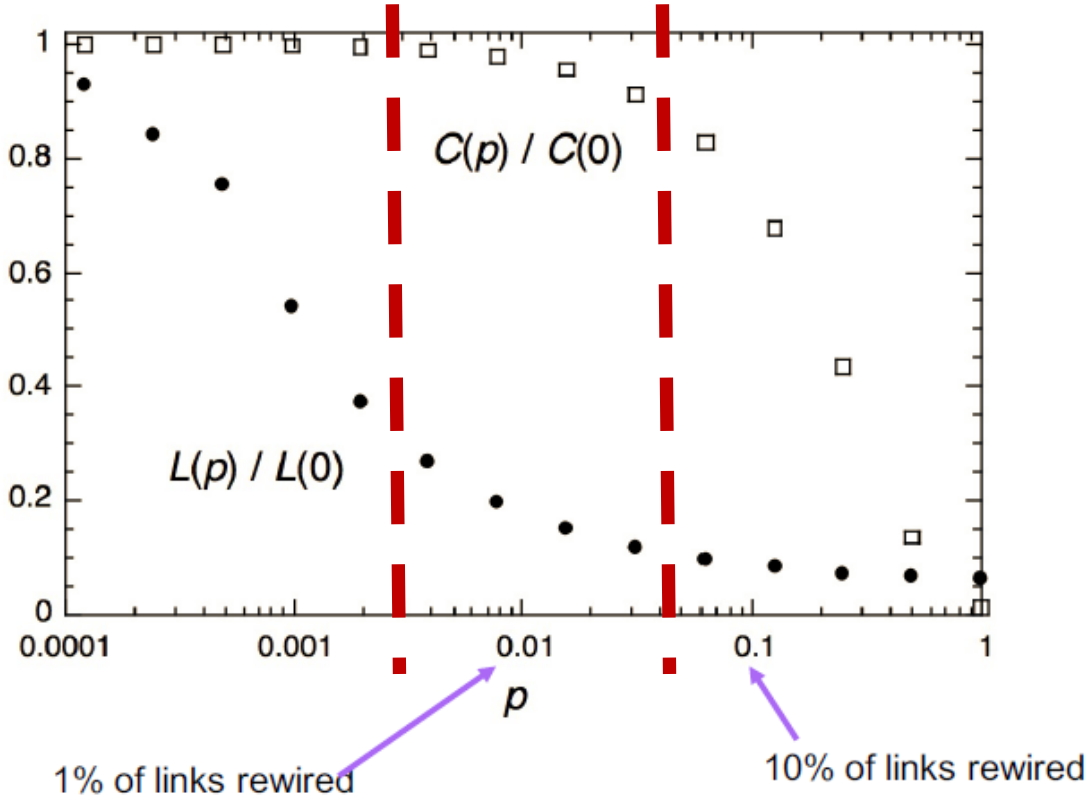
- Erdos-Renyi networks can grow to be very large but nodes will be just a few steps apart




$$l_{av} \sim \frac{\log N}{\log z}$$

# As we increase p from 0 to 1

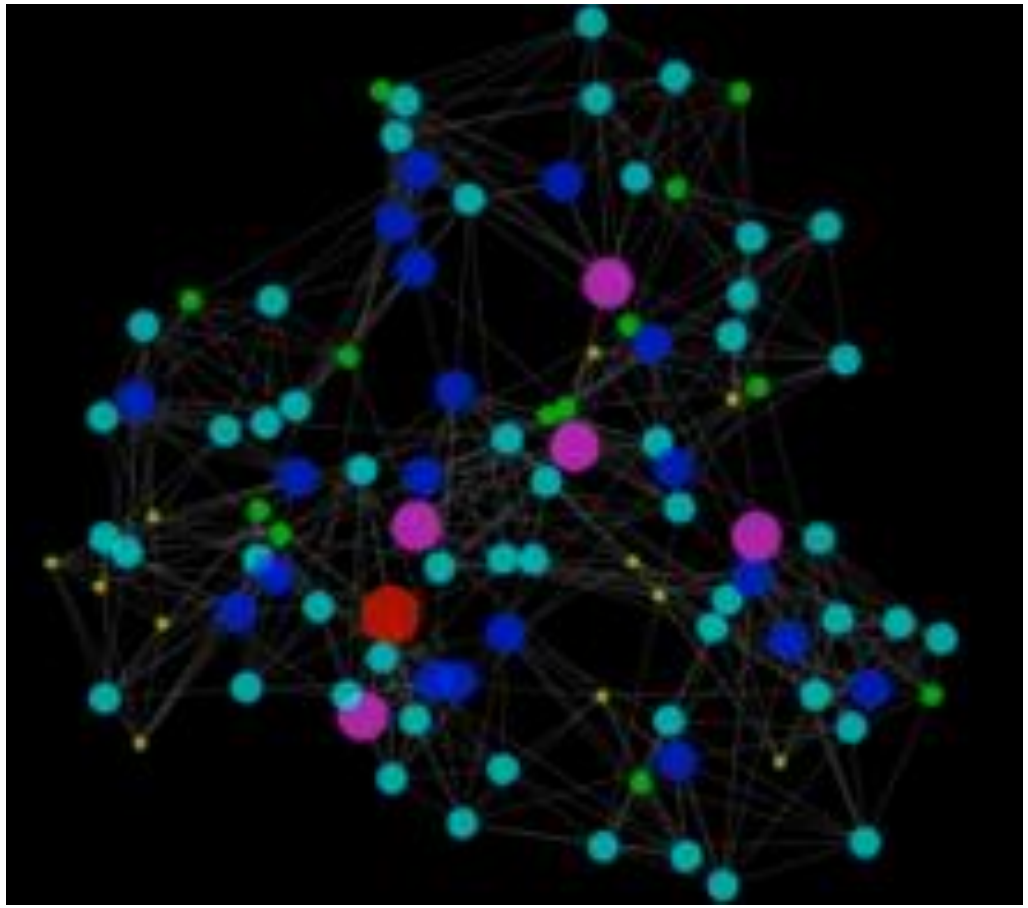



Source: Watts, D.J., Strogatz, S.H.(1998) Collective dynamics of 'small-world' networks. Nature 393:440-442.

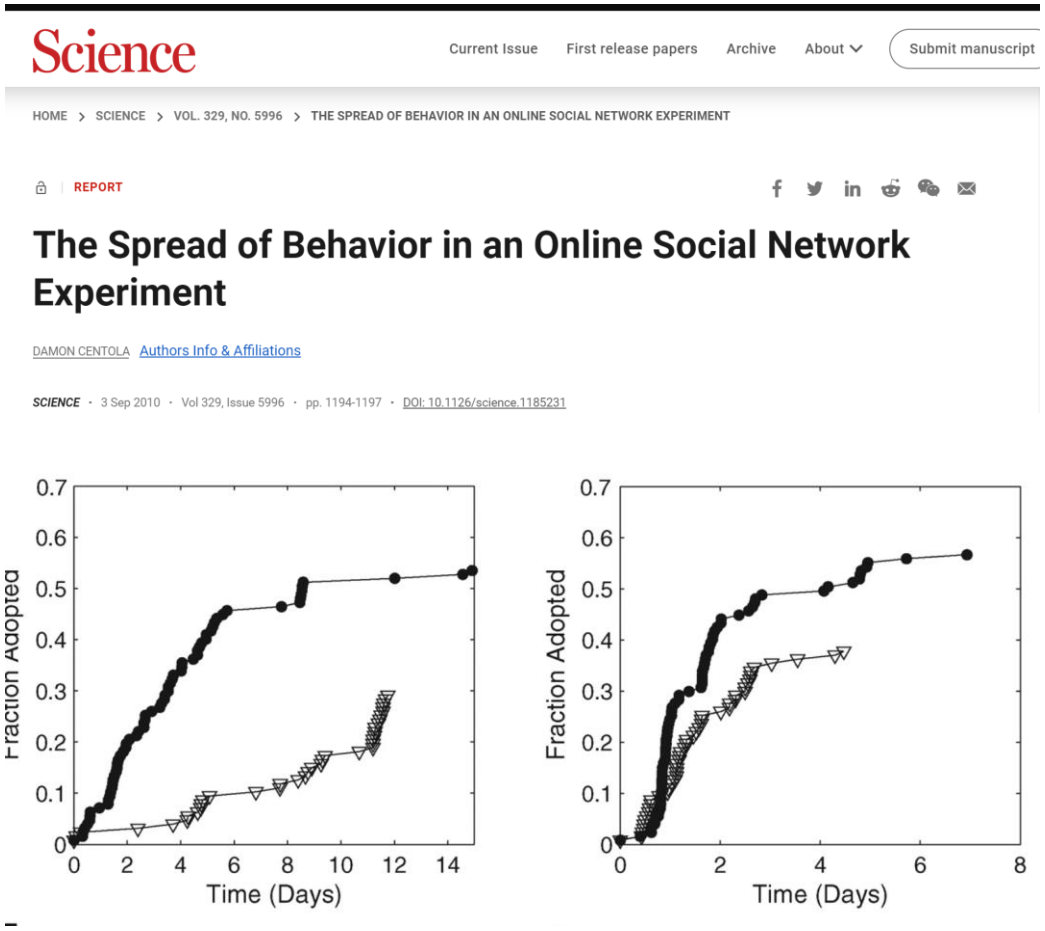
- Fast decrease of mean distance, i.e., ASP
- Slow decrease in clustering

The Watts–Strogatz model is a random graph generation model that produces graphs with small-world properties, including short average path lengths and high clustering.

# Watts-Strogatz model visualized



# Implications for social media influences





## Abstract

How do social networks affect the spread of behavior? A popular hypothesis states that networks with many clustered ties and a high degree of separation will be less effective for behavioral diffusion than networks in which locally redundant ties are rewired to provide shortcuts across the social space. A competing hypothesis argues that when behaviors require social reinforcement, a network with more clustering may be more advantageous, even if the network as a whole has a larger diameter. I investigated the effects of network structure on diffusion by studying the spread of health behavior through artificially structured online communities. Individual adoption was much more likely when participants received social reinforcement from multiple neighbors in the social network. The behavior spread farther and faster across clustered-lattice networks than across corresponding random networks.

# Limitation of WS model

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- Does not take into consideration of geography – long rang links not as likely as short range ones
- Does not take into consideration of hierarchical structure/groups
- No hubs

# Lab 3

- WS model construction
- In-class exercise